

Sleep Posture Recognition via Conventional Machine Learning, Pose-Guided Feature Engineering, and Deep Transfer Learning: A Comparative Analysis

Jeevan Jyoti Dash

*School of Computer Science and Engineering
Vellore Institute of Technology
Chennai, Tamil Nadu, India
jeevanjyoti.dash2022@vitstudent.ac.in*

Dr. Omana J

*School of Computer Science and Engineering
Vellore Institute of Technology
Chennai, Tamil Nadu, India
omana.j@vit.ac.in*

Abstract—Body position during sleep directly influences rest quality, physical well-being, and stress levels, making non-contact automated posture detection a meaningful asset in contemporary health surveillance systems. While image-based monitoring eliminates the inconvenience and light sensitivity associated with wearable sensors, it remains uncertain how much representational complexity such a system genuinely requires: earlier approaches typically fix on either manually crafted pose geometry or deep networks, without quantifying the performance gain at each intermediate step, giving practitioners little basis on which to allocate modelling effort when working with small, domain-specific imaging collections. This work addresses that question through a single, tightly controlled sequence of six representation strategies applied to 455 infrared selective (IRS) sleep photographs covering five body positions (Foetus, Log, Starfish, Unknown, Yearner): manually engineered geometric descriptors from YOLOv11 pose keypoints fed into Random Forest, Extra Trees, SVM, XGBoost, and CatBoost classifiers; a Q-Learning agent that conducts an autonomous search for a discriminative feature subset; a lightweight collection of pose-only measurements fed into XGBoost; ResNet50 and EfficientNetB3 trained via transfer learning; a fusion architecture merging deep and pose features; and ConvNeXt-Tiny fine-tuned on a synthetically enlarged dataset. Evaluated through 5-fold cross-validation and an independent held-out set, the manually engineered feature classifiers saturated between 21.5% and 29.0% accuracy, the pose-only approach attained 42.3%–43.3%, ResNet50 and EfficientNetB3 attained 63.7% and 62.0% respectively, the fusion model reached 56.7%, and ConvNeXt-Tiny with augmentation achieved the highest result of 80.55% accuracy with a weighted F1-score of 0.80, establishing that end-to-end learned visual features substantially outperform handcrafted or automatically selected geometric representations on this task.

Keywords—Sleep Posture Classification, Deep Learning, ConvNeXt, Transfer Learning, YOLOv11 Pose Estimation, Q-Learning, Reinforcement Learning, Feature Selection, Computer Vision, Infrared Imaging, Healthcare Analytics.

I. Introduction

Sleep is an active physiological process. The bodily orientation sustained across a full night determines how refreshed, comfortable, and physiologically regulated a person feels on waking. Uneven mechanical loading of the joints or vertebral column over many hours can result in morning stiffness, disrupted sleep continuity, and quantifiable increases in stress. Identifying the actual sleeping position of an individual in a non-intrusive manner therefore offers tangible benefit for both clinical practice and consumer wellness applications.

Sensor-driven solutions have traditionally represented the mainstream in this domain. Sensors embedded within pillows and wearable devices have been deployed to estimate stress and infer sleeping habits [1], [4], and flexible body-attached sensors have been used to establish statistical links between sleeping orientation and independently measured sleep quality [2]. Although these approaches are effective, they share a fundamental limitation: the sensing hardware interposes between the body and the sleeping surface, potentially modifying the very behaviour under observation. Image-based methods circumvent this problem. Infrared selective (IRS) cameras record a sleeping subject's silhouette without requiring any physical contact, tolerate the near-total darkness typical of a bedroom environment, and have already been coupled with convolutional architectures [7] and ambient monitoring platforms [10] to enable visually driven posture recognition.

What the existing literature does not yet resolve is how much representational complexity this recognition task truly demands. Certain pipelines depend exclusively on geometric analysis of detected skeletal joints; others circumvent joint detection entirely and instead learn directly from raw pixel data through a deep network. Rarely are these strategies evaluated under identical experimental conditions, leaving an open practical question for system designers: is a relatively simple keypoint-based pipeline sufficient for a realistically sized dataset, or does achieving competitive performance genuinely require the engineering overhead and computational burden of a deep convolutional backbone?

Problem Statement: For a compact, infrared-captured, five-class sleep posture collection (Foetus, Log, Starfish, Unknown, Yearner), it has not yet been established which representational strategy produces the most dependable classifier and by what magnitude it surpasses alternatives, whether the winning approach is manually designed joint geometry, an autonomously chosen subset of that geometry, a condensed pose summary, or a fully learned visual feature set.

Objectives: The goals of this investigation are: (1) to construct a unified infrared sleep-posture processing pipeline encompassing keypoint detection, coordinate normalization, and feature computation; (2) to assess the performance of established machine learning classifiers on manually engineered geometric descriptors; (3) to determine whether a Q-Learning feature-selection agent can surpass the manually engineered baseline; (4) to evaluate a streamlined set of pose-only measurements; (5) to fine-tune ResNet50, EfficientNetB3, and ConvNeXt-Tiny on raw images, the last employing aggressive data augmentation; and (6) to merge deep and manual features within a hybrid architecture, enabling all six strategies to be compared under a uniform protocol measuring accuracy, precision, recall, and F1-score.

Novelty: The primary methodological contribution of this paper is the **Q-Learning-based feature-selection agent** presented as Algorithm 1 in Section III-F, which, to the best of our knowledge, has not previously been applied to search the handcrafted pose-geometry feature space for sleep-posture recognition. Paired with a single, controlled, six-stage comparison spanning handcrafted features, automated feature selection, pose-only descriptors, deep transfer learning, and deep-pose hybrid fusion, all evaluated on one shared infrared dataset, this provides what is, to our knowledge, the first staged account of how much accuracy each added layer of representational complexity is actually worth for infrared sleep-posture classification.

The remainder of this paper proceeds as follows. Section II reviews prior research on sleep posture recognition and stress monitoring. Section III describes the dataset, the overall system design, and each of the six modelling approaches, along with the evaluation framework used for comparison. Section IV details the experimental configuration and presents accuracy, precision, recall, and F1-score results for all strategies, supported by confusion matrices and training progression plots. Section V concludes the paper with a synthesis of findings and directions for future investigation.

II. Related Works

One research direction approaches sleep-related stress as a sensing challenge addressed at the bedding interface. Rachakonda et al. integrated sensors into a standard pillow to identify stress markers from nocturnal behaviour [1], subsequently expanding this concept into SayoPillow, an IoMT architecture incorporating blockchain-based data privacy into the same measurement pipeline [4]. Zhang et al. explored a complementary path, affixing a flexible sensor to the body and quantitatively linking recorded body orientation to separately evaluated sleep quality [2]. Reviews by Nath et al. [3], [11] and Akhtar et al. [5] position such sensor-based work within the broader domain of machine-learning-driven stress assessment, while Rahman et al. reframe the same challenge using logged sleep-activity records rather than direct sensor outputs [6].

A parallel line of inquiry relies on visual data rather than embedded hardware. Tang et al. applied a convolutional neural network trained on captured video frames to identify sleeping postures [7], while Dhamchatsoontree et al. fused visual and behavioural cues within an “i-sleep” platform designed to characterise overall sleep patterns beyond a single positional label [8]. Febriana et al. employed a Microsoft Kinect depth-sensing device to monitor nocturnal body movement without physical contact [18], and Barsocchi et al. developed a non-intrusive camera-based monitoring arrangement to facilitate sleep behaviour research [10]. Jayawickrama and Rupasingha presented two complementary contributions: a rule-and-machine-learning hybrid for stress inference from sleeping habits [9], and an ensemble-based strategy applied to the same problem using sleep-phase behavioural data [13].

A third stream furnishes the foundational tooling on which both preceding groups depend. Widely adopted computer vision frameworks such as OpenCV [15] and compact perception pipelines such as MediaPipe [16] are routinely applied to extract structured keypoint data from images, while the comprehensive survey by Zheng et al. traces the maturation of deep-learning-based human pose estimation as a discipline [17]. Beyond the sleep context, comparable sensing concepts have been applied

to stress inference from seated foot-pressure patterns [19] and from physiological and behavioural signals in occupational environments [12]. Clinical investigations have additionally explored the relationship between infant sleep orientation and autonomic function as well as arousal thresholds [20], and wearable devices have been used to monitor the physiological consequences of stress on the sleep of pilgrims during prolonged journeys [14].

Collectively, these studies demonstrate that sensor-based and vision-based architectures alike can capture meaningful information about nocturnal behaviour, and that well-established pose estimation frameworks are already available for deriving structured landmark data from sleep imagery. However, none of them presents a rigorous, controlled evaluation of the accuracy change at each transition between a fully manual geometric pipeline and an end-to-end learned deep visual pipeline on a shared dataset. Filling this gap, by means of the novel feature-selection agent and the incremental comparison outlined above, is the central aim of the present study.

III. Methodology

The experimental framework employed in this study begins with a common preprocessing phase comprising infrared image acquisition, YOLOv11-based keypoint detection, and landmark normalisation, followed by four independent modelling pathways: a manual geometric-feature pathway, a compact pose-descriptor pathway, a deep transfer-learning pathway, and a hybrid pathway that combines the deep and pose-descriptor outputs. Fig. 2 depicts this overall workflow; the subsequent subsections describe the dataset, the system structure, and each individual pathway.

A. Dataset Description

The dataset consists of 455 infrared selective (IRS) photographs, each depicting a sleeping individual in one of five body position categories: Foetus, Log, Starfish, Unknown, and Yearner. Table I provides the per-class image counts. Class sizes are not uniformly distributed; Log constitutes the largest group at 126 samples while Starfish is the smallest at 71. This distribution asymmetry is addressed throughout the study via stratified partitioning for both cross-validation and train-validation splitting, and by reporting weighted-average metrics in addition to per-class breakdowns.

Posture Class	Images	Share (%)
Foetus	96	21.1
Log	126	27.7
Starfish	71	15.6
Unknown	86	18.9
Yearner	76	16.7
Total	455	100.0

TABLE I. CLASS DISTRIBUTION OF THE IRS SLEEP POSTURE DATASET

Every image is a single-channel infrared frame in which the recumbent individual appears as a luminous outline against a dark field, with bed coverings regularly concealing sections of the limbs. This characteristic motivates the keypoint imputation procedure presented in Section III-C. Fig. 1 displays ten representative examples from the Unknown category, highlighting the within-class visual variability that classifiers are required to handle.

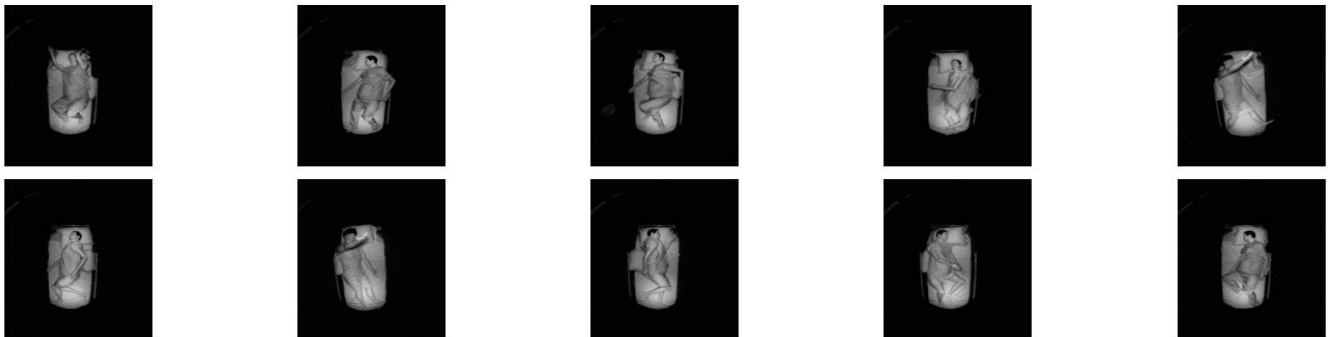


Fig. 1. Ten representative infrared selective (IRS) sleep images drawn from the “Unknown” posture class.

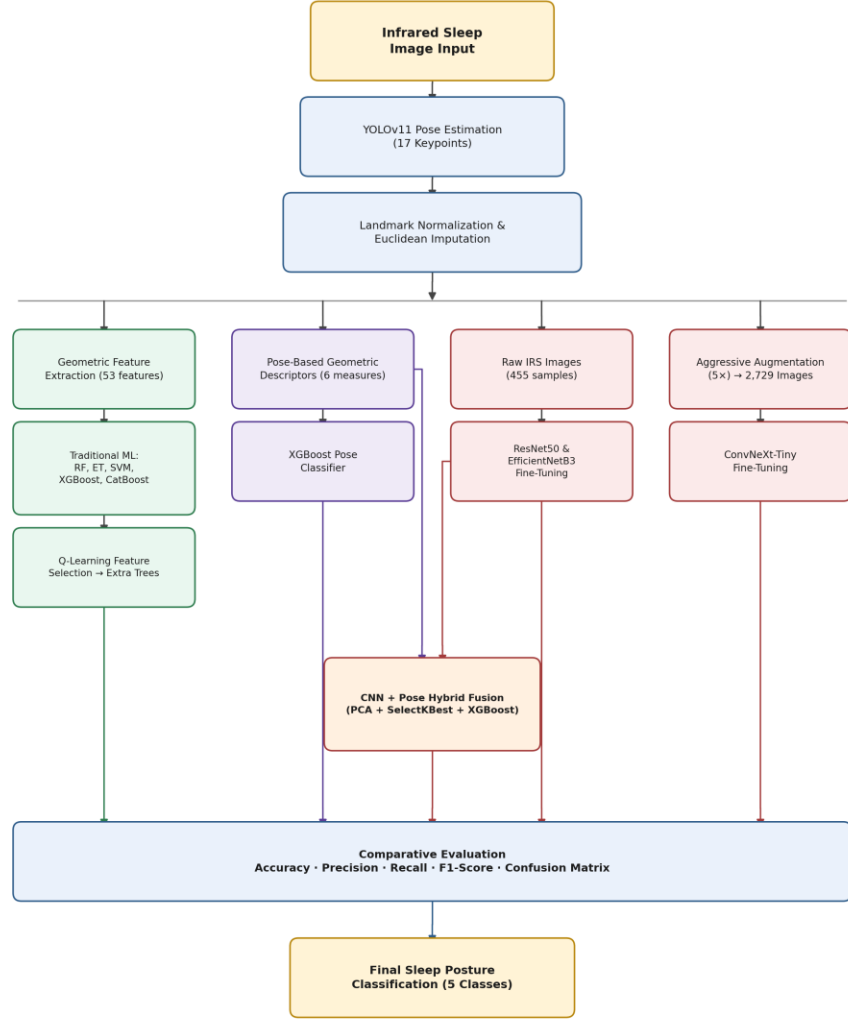


Fig. 2. Overall system architecture: shared preprocessing, four parallel representation branches, the CNN + Pose hybrid fusion node, and the comparative evaluation stage.

B. Pose Landmark Detection

Skeletal keypoint extraction was performed using YOLOv11 in its pose-estimation mode (YOLO11n-pose) [17], which identifies 17 canonical landmark positions per frame: the nose, eyes, ears, shoulders, elbows, wrists, hips, knees, and ankles. For the present study, the 12 limb and torso keypoints (shoulders, elbows, wrists, hips, knees, ankles) were retained for feature construction, as they encode the primary discriminative information needed to distinguish the Foetus, Log, Starfish, Unknown, and Yearner postures, while the facial keypoints contribute relatively little. Fig. 3 illustrates the detector output superimposed on a representative IRS frame, including the bounding box used for downstream coordinate normalisation.

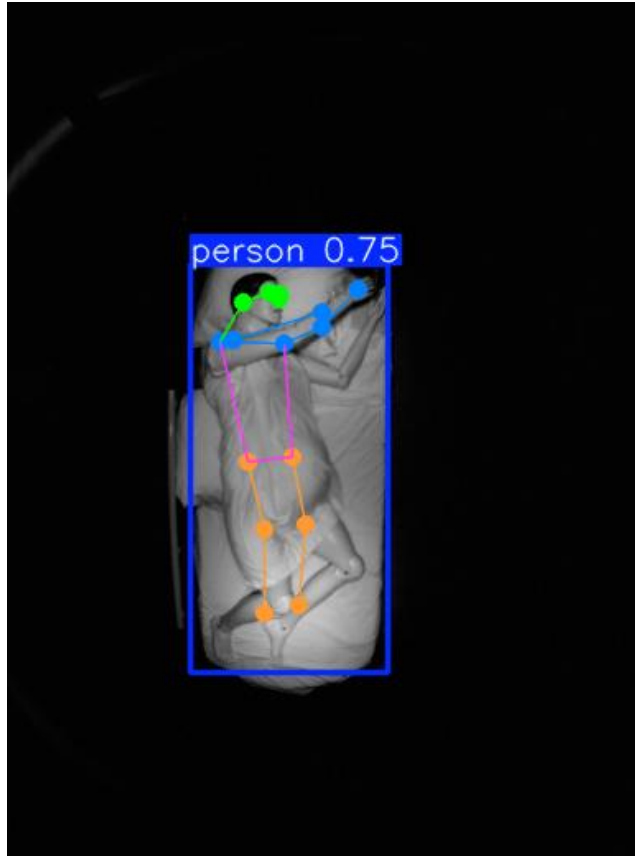


Fig. 3. YOLOv11 pose detection on an IRS sleep image, with the bounding box and skeletal keypoints used for subsequent feature extraction.

C. Missing Value Imputation

Bed coverings often occlude one or several joints from the camera's field of view, causing a notable proportion of keypoint coordinates to arrive as missing values rather than merely as noisy observations. Each incomplete data row was reconstructed by identifying its nearest neighbour within the remaining rows and substituting the absent values from that neighbour. Distance between two coordinate rows p and q , computed over whichever dimensions were available in both, was measured using the standard Euclidean form

$$d(p, q) = \sqrt{\sum_i (p_i - q_i)^2} \quad (1)$$

This neighbour-finding and value-imputation procedure was applied across five sequential passes, allowing values inserted in one pass to assist in resolving residual gaps in subsequent ones. Any coordinate still absent following five passes was replaced with the corresponding column mean, thereby ensuring a fully populated landmark matrix prior to feature construction.

D. Geometric Feature Engineering

From the cleaned landmark matrix, an initial set of 41 geometric descriptors was derived for each image: inter-keypoint distances between anatomically relevant pairs, joint flexion angles at the knees, hips, and elbows, the horizontal and vertical extents of the keypoint distribution, bilateral asymmetry differences, and trigonometric encodings of each joint angle. Each joint angle θ was computed at a vertex B given its two neighbouring landmarks A and C , by forming the vectors $v1 = A - B$ and $v2 = C - B$ and applying

$$\theta = \arccos((v1 \cdot v2) / (\|v1\| \|v2\|)) \quad (2)$$

Six unitless ratio features (including hip-to-shoulder width ratio and left-to-right limb length ratios) and six degree-based re-encodings of the joint angles were subsequently appended, expanding the feature vector to 53 dimensions. Table II categorises these features by type.

Category	Representative Features	Count
Distances	shoulder_width, hip_width, shoulder-hip dist., hip-knee dist., knee-ankle dist., torso_diagonal	15
Joint Angles	left/right knee, hip, and elbow angles (Eq. 2)	6
Spread	horizontal_spread, vertical_spread, spread_ratio	3
Symmetry	shoulder, hip, knee, ankle, elbow L-R differences	5
Trigonometric	sine and cosine of each of the 6 joint angles	12
Derived Ratios	hip-shoulder, L/R leg, L/R arm, torso-shoulder, torso-hip, spread_ratio_2	6
Angle-in-Degrees	knee/hip/elbow angles via arctan2(sin, cos)	6
Total		53

TABLE II. GEOMETRIC FEATURE SET BY CATEGORY (53 FEATURES)

E. Traditional Machine Learning Classification

The 53-dimensional feature vectors were label-encoded and standardised prior to being supplied to five established classifiers: Random Forest, Extra Trees, an RBF-kernel SVM, XGBoost, and CatBoost, each assessed through 5-fold stratified cross-validation. Table III documents the hyperparameter settings adopted for each model; corresponding accuracy figures are reported in Section IV.

Model	Key Hyperparameters
Random Forest	n_estimators = 500, random_state = 42
Extra Trees	n_estimators = 500, random_state = 42
SVM (RBF)	kernel = rbf, C = 10, gamma = scale
XGBoost	n_estimators = 500, max_depth = 6, lr = 0.05, subsample = 0.8, colsample_bytree = 0.8
CatBoost	iterations = 500, depth = 6, learning_rate = 0.05

TABLE III. HYPERPARAMETERS OF THE TRADITIONAL MACHINE LEARNING CLASSIFIERS

F. Reinforcement-Learning-Based Feature Selection (Novel Contribution)

This subsection introduces the principal methodological contribution of the paper. **To determine whether a more targeted subset drawn from the 41 base features could surpass the full-feature baseline, a Q-Learning agent—novel within this feature-engineering context—was developed to explore the feature subset space directly**, following the procedure outlined in **Algorithm 1**. The agent's state is represented as a binary mask across the 41 features; its action toggles a single feature into or out of the current mask; and its reward is the 3-fold cross-validation accuracy of a rapid 50-tree Extra Trees classifier evaluated on the active feature subset. After taking action a in state s and observing reward r and next state s' , the agent's Q-table was updated using the standard tabular Q-learning rule

$$Q(s,a) \leftarrow Q(s,a) + \alpha [r + \gamma \max_{a'} Q(s',a') - Q(s,a)] \quad (3)$$

The agent completed 60 episodes using a learning rate of $\alpha = 0.1$, a discount factor of $\gamma = 0.9$, and an exploration rate ϵ decaying linearly from 1.0 to 0.1 over the course of training, with a ceiling of 20 toggle steps per episode. Algorithm 1 outlines the complete search procedure.

NOVEL CONTRIBUTION

Algorithm 1: Q-Learning Feature Selection

```

Input: feature matrix X (n x 41), labels y, episodes=60
Initialize Q-table Q[state][action] = 0 for all states
epsilon = 1.0; decay = (1.0-0.1)/episodes
for episode = 1 to episodes:
    mask = random binary vector of length 41
    for step = 1 to min(41, 20):
        state = bitstring(mask)
        if rand() < epsilon: action = random feature index
        else: action = argmax_a Q[state][a]

```

```

new_mask = toggle(mask, action)
reward = 3-fold CV accuracy of ExtraTrees(50)
    trained on X[:, new_mask]
next_state = bitstring(new_mask)
Q[state][action] += alpha * (reward
    + gamma * max_a' Q[next_state][a']
    - Q[state][action])
mask = new_mask
track best_reward, best_subset
epsilon = max(0.1, epsilon - decay)
return best_subset, best_reward, reward_history

```

Fig. 4 plots the per-episode reward throughout training.

The curve fluctuates in a narrow band, roughly between 0.21 and 0.27, across all 60 episodes without demonstrating a sustained upward trend. The strongest subset identified by the agent, comprising 19 of the original 41 features, was already discovered by the tenth episode and was never subsequently improved upon, stabilising at a cross-validation accuracy of 29.0%. Section IV discusses this convergence plateau at greater length; in essence, an automated search is unable to produce discriminative information that was not inherent in the underlying manually constructed feature pool.

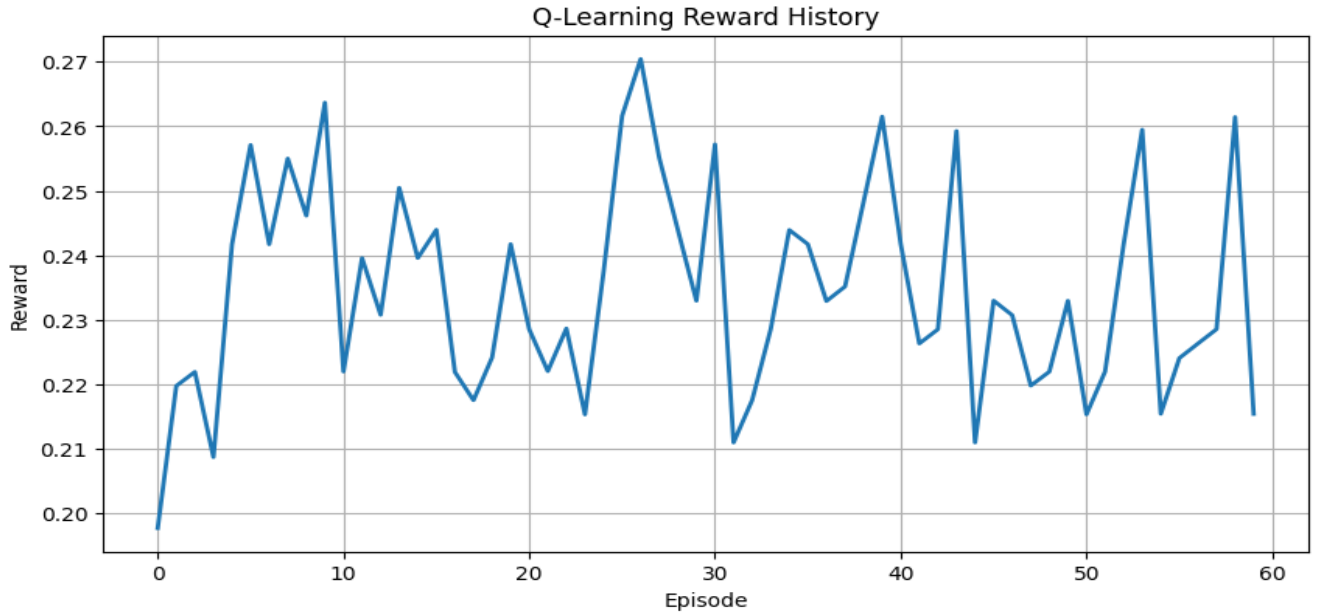


Fig. 4. Q-Learning reward (3-fold cross-validation accuracy of the search policy) over 60 episodes, plateauing by episode 10.

G. Pose-Descriptor-Based Classification

As a more compact alternative to the 53-dimensional manually engineered vector, a second pipeline derived only six global body measurements directly from the raw YOLOv11 keypoints: shoulder breadth, hip breadth, left and right torso length, left and right arm length, left and right leg length, and total body height. These were fed into XGBoost (300 estimators, maximum depth 6, learning rate 0.05) under the same 5-fold stratified evaluation protocol. Despite its considerably smaller dimensionality, this descriptor collection outclassed every classifier relying on the larger manually crafted vector, a finding elaborated in Section IV.

H. Deep Transfer Learning

Two convolutional backbones underwent fine-tuning directly on the original, unaugmented collection of 455 images. ResNet50 was initialised with ImageNet pre-trained weights and extended with global average pooling, a 512-unit ReLU fully connected layer, a dropout layer with rate 0.3, and a five-class softmax output; training proceeded for 20 epochs with the backbone weights frozen, followed by 15 additional epochs in which the final 50 backbone layers were unfrozen and updated at a reduced learning rate of 1×10^{-5} . EfficientNetB3 adopted a comparable training schedule at a higher input resolution of 300×300 pixels with a batch size of 16, incorporating early stopping (patience 5 on validation accuracy) and learning-rate reduction on plateau (scale factor 0.2, patience 3). Table IV summarises the configuration adopted for both models along with ConvNeXt-Tiny, which is described in Section III-J.

Model	Input	Batch	Training Regime	Optimizer
ResNet50	224×224	32	20 epochs frozen + 15 fine-tune (last 50 layers)	Adam
EfficientNetB3	300×300	16	Early stop (pat. 5) + LR reduce on plateau	Adam
ConvNeXt-Tiny	224×224	32	Frozen backbone, 8 epochs, early stop + LR reduce	Adam

TABLE IV. DEEP TRANSFER LEARNING MODEL CONFIGURATIONS

I. CNN + Pose Hybrid Fusion

To investigate whether deep visual representations and manually engineered pose geometry encode complementary information, a fusion model was constructed by extracting ResNet50's deep features for each image, reducing their dimensionality with PCA (preserving 95% of total variance), and concatenating the compressed result with the six standardised pose descriptors from Section III-G. Mutual-information-based feature selection (SelectKBest, $k = 100$) was then applied to the combined vector, which was subsequently classified using XGBoost (1000 estimators, maximum depth 10, learning rate 0.01, subsample rate 0.8, column subsampling rate 0.8) under 5-fold stratified cross-validation.

J. Data Augmentation and ConvNeXt-Tiny Fine-Tuning

Given that 455 images represent a relatively limited sample for deep backbone fine-tuning, a data augmentation procedure was introduced specifically for the final model in this investigation. Each source image was synthetically expanded into five additional variants through random rotation ($\pm 25^\circ$), width and height translation ($\pm 20\%$), zoom variation ($\pm 20\%$), horizontal flipping, and brightness perturbation ($0.8-1.2\times$). This expanded the working dataset from 455 to 2,729 images, divided into 2,184 for training and 545 for validation whilst preserving the original 80/20 partition. ConvNeXt-Tiny, initialised with ImageNet pre-trained weights, was then fine-tuned on this enlarged set using a frozen backbone coupled with global average pooling, batch normalisation, a 512-unit GELU dense layer, a dropout layer at rate 0.5, and a five-class softmax head, optimised with Adam and subject to early stopping (patience 5 on validation accuracy) and learning-rate reduction on plateau (scale factor 0.2, patience 3) for a maximum of 8 epochs.

K. Performance Evaluation Metrics

Regardless of their internal design, all classifiers in this study ultimately generate a probability vector over the five posture classes for each input x . For the neural network models this distribution is obtained by passing the final layer's raw scores z through a softmax,

$$\hat{y}_c = \exp(z_c) / \sum_k \exp(z_k) \quad (4)$$

and training proceeds by minimizing the categorical cross-entropy between this predicted distribution and the one-hot ground truth across all N training samples,

$$L = -(1/N) \sum_i \sum_c y_{i,c} \log(\hat{y}_{i,c}) \quad (5)$$

For the ensemble and gradient-boosted classifiers used in the handcrafted and pose-based branches, the analogous multi-class log-loss and Gini/entropy split criteria serve the same purpose internally. Once trained, every model in this study is judged on the same four metrics, together with the confusion matrix:

$$Accuracy = (TP + TN) / (TP + TN + FP + FN) \quad (6)$$

$$Precision = TP / (TP + FP) \quad (7)$$

$$Recall = TP / (TP + FN) \quad (8)$$

$$F1 = 2 \cdot (Precision \cdot Recall) / (Precision + Recall) \quad (9)$$

where TP, TN, FP, and FN are the true positive, true negative, false positive, and false negative counts, computed per class and combined by weighted averaging across the five posture classes to account for the class imbalance noted in Table I.

IV. Results and Discussion

A. Experimental Setup

All six approaches were implemented in Python: scikit-learn handled the conventional classifiers and cross-validation utilities, XGBoost and CatBoost provided gradient-boosted model implementations, the Ultralytics library supplied the YOLOv11 pose detection backbone, and TensorFlow/Keras supported ResNet50, EfficientNetB3, and ConvNeXt-Tiny. A global random seed of 42 was applied at every point requiring a stochastic initialisation or data split, ensuring full reproducibility of the reported figures. The conventional, pose-based, and hybrid classifiers were evaluated via 5-fold stratified cross-validation; the three deep transfer-learning models were assessed on a held-out 20% stratified validation partition, consistent with standard fine-tuning practice. All training was carried out on a GPU-enabled cloud environment. Each ConvNeXt-Tiny epoch on the augmented 2,184-image training split required approximately 67 to 70 minutes, meaning the complete 8-epoch training run demanded roughly nine hours of GPU computation, compared with sub-second per-fold costs for the conventional and pose-based classifiers. This practical computational disparity is an important consideration when interpreting the accuracy results that follow.

B. Comparative Accuracy Across All Methods

Table V and Fig. 5 consolidate the accuracy figures across all nine evaluated configurations, arranged to reflect the incremental progression outlined in Section III: manually engineered features, reinforcement-learning-driven feature selection, pose-only descriptors, deep transfer learning, hybrid feature fusion, and augmentation-assisted transfer learning.

TABLE V. COMPARATIVE ACCURACY ACROSS ALL EVALUATED METHODS

Stage	Method	Accuracy
Traditional ML	Random Forest	27.0%
Traditional ML	XGBoost	26.0%
Traditional ML	CatBoost	26.0%
RL Feature Selection	Q-Learning (Extra Trees)	29.0%
Pose-Based Features	YOLO Pose + XGBoost	43.3%
Deep Transfer Learning	ResNet50	63.7%
Deep Transfer Learning	EfficientNetB3	62.0%
Hybrid Fusion	CNN + Pose Hybrid	56.7%
Deep TL + Augmentation	ConvNeXt-Tiny	80.55%

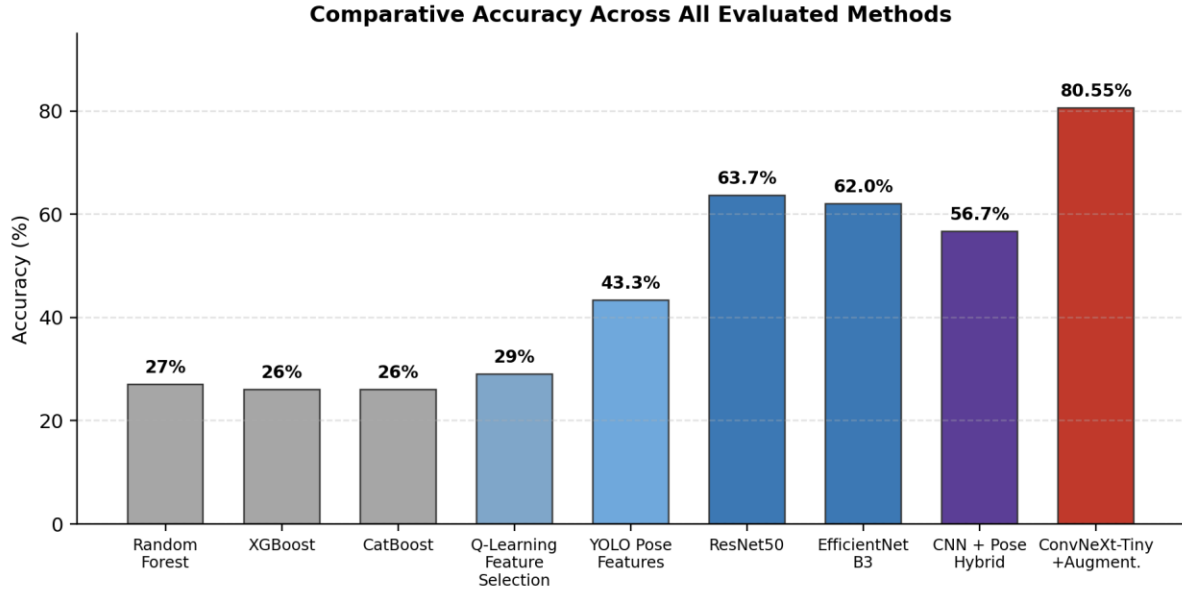


Fig. 5. Accuracy achieved by all nine evaluated methods, showing the staged climb from handcrafted machine learning through pose-based features to deep transfer learning, with ConvNeXt-Tiny and augmentation in front.

C. Handcrafted Feature Classifiers and Feature Selection

The three manual-feature classifiers highlighted in Table V fall within a single percentage point of each other, clustering between 26% and 27%. Table VI supplies results for the two classifiers omitted from that summary comparison, Extra Trees and an RBF SVM, providing the complete evaluation across all five models assessed on the 53-dimensional feature vector.

Classifier	5-Fold CV Accuracy
Random Forest	27.25%
Extra Trees	28.13%
Support Vector Machine (RBF)	21.54%
XGBoost	26.59%
CatBoost	26.37%

TABLE VI. COMPLETE TRADITIONAL MACHINE LEARNING BENCHMARK (53 FEATURES)

Five architecturally distinct classifiers—a tree ensemble, a margin-based SVM, and two gradient-boosting variants—all fall within a 6.6-percentage-point window (21.5%–28.1%), implicating the feature space itself, rather than any individual classifier's inductive preference, as the primary performance ceiling. The Q-Learning agent from Section III-F was explicitly designed to challenge this interpretation: if a superior subset of the same 41 base features existed, a systematic search should have located it. The agent did yield marginal improvement, but only to 29.0%, and as described earlier, this peak was already attained by the tenth episode out of sixty and remained unchanged thereafter. This pattern indicates that the search had exhausted the discriminative content of the feature pool, rather than simply running out of time to explore it; and this constitutes the clearest evidence in this study that Algorithm 1, though a genuine methodological addition, cannot extract performance from manual features that were never sufficiently informative to begin with.

The pose-only descriptor pipeline presents a more optimistic narrative for compact feature representations. Table VII documents the per-fold accuracy of the six-measurement XGBoost classifier outlined in Section III-G.

CV Fold	Accuracy
Fold 1	39.51%
Fold 2	45.68%
Fold 3	41.25%
Fold 4	45.00%

CV Fold	Accuracy
Fold 5	40.00%
Mean	42.29%

TABLE VII. YOLO POSE DESCRIPTOR 5-FOLD CROSS-VALIDATION SCORES

Despite having approximately nine times fewer dimensions than the manually engineered vector, this compact descriptor set surpassed every manual-feature classifier by a clear margin. A credible explanation lies in the sample-to-feature ratio: distributing 455 images across 53 manually crafted dimensions yields fewer than nine training examples per dimension, an unfavourable regime for tree and margin-based learning algorithms. In contrast, the same 455 images spread across only six pose measurements provide more than seventy examples per dimension, supplying sufficient data for the classifier to learn well-supported decision boundaries rather than memorising noise.

D. Deep Transfer Learning Models

Direct fine-tuning on raw pixel data yielded the most substantial single-step improvement across the entire experimental sequence. ResNet50 attained 63.7% accuracy and EfficientNetB3 reached 62.0%, each approximately twenty percentage points above the top pose-based result and more than twice the accuracy of any manually engineered feature classifier. Tables VIII and IX provide per-class breakdowns of these summary figures.

Class	Precision	Recall	F1-Score	Support
Foetus	0.83	0.83	0.83	18
Log	0.60	0.60	0.60	25
Starfish	0.70	0.50	0.58	14
Unknown	0.55	0.52	0.54	21
Yearner	0.56	0.77	0.65	13

TABLE VIII. RESNET50 PER-CLASS METRICS (OVERALL ACCURACY 63.7%)

Class	Precision	Recall	F1-Score	Support
Foetus	0.60	0.83	0.70	18
Log	0.68	0.60	0.64	25
Starfish	0.87	0.93	0.90	14
Unknown	0.60	0.29	0.39	21
Yearner	0.37	0.54	0.44	13

TABLE IX. EFFICIENTNETB3 PER-CLASS METRICS (OVERALL ACCURACY 62.0%)

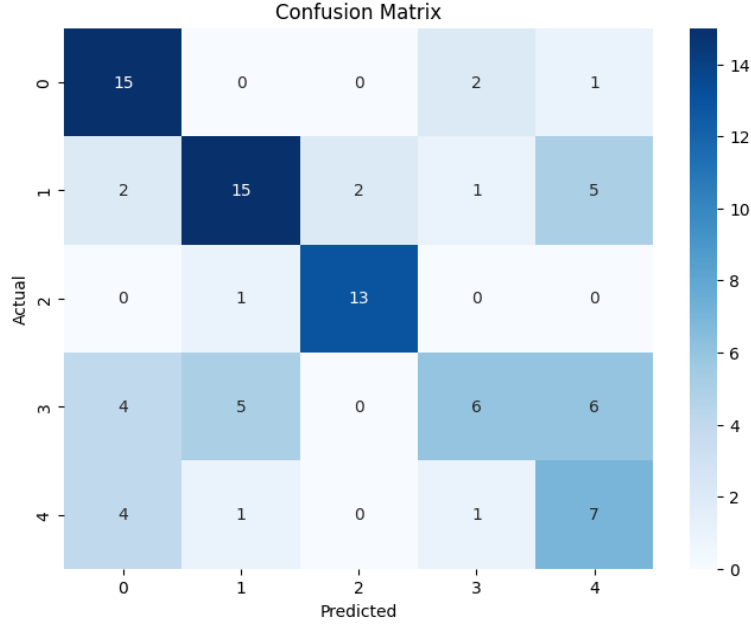


Fig. 6. EfficientNetB3 confusion matrix on the validation split (class order: Foetus, Log, Starfish, Unknown, Yearner).

The two backbones achieve comparable overall accuracy via complementary per-class strengths: ResNet50 performs best on the Foetus class ($F1 = 0.83$) and least reliably on Unknown ($F1 = 0.54$), whereas EfficientNetB3 excels at Starfish ($F1 = 0.90$, recall 0.93) yet underperforms on Unknown ($F1 = 0.39$) and Yearner ($F1 = 0.44$). Given that the two networks commit different errors on essentially the same data, an ensemble combining their outputs—not examined here—represents an obvious avenue for further improvement. In any case, both results point to the same conclusion: raw image texture and boundary information, which a keypoint-only pipeline discards entirely, clearly encodes posture-discriminative signal that skeletal geometry alone fails to capture, particularly when bedding occludes portions of the body.

E. Hybrid Fusion of Deep and Pose Features

Merging ResNet50’s deep feature representations with the pose descriptors, as detailed in Section III-I, yielded a mean accuracy of 56.7%; Table X provides the per-fold breakdown.

CV Fold	Accuracy
Fold 1	64.20%
Fold 2	49.38%
Fold 3	56.25%
Fold 4	52.50%
Fold 5	61.25%
Mean	56.72%

TABLE X. CNN + POSE HYBRID FUSION 5-FOLD CROSS-VALIDATION SCORES

The nearly 15-percentage-point spread between the weakest fold (49.4%) and the strongest (64.2%) suggests representational instability in the merged feature vector once it undergoes compression for a gradient-boosted classifier. More tellingly, the fusion approach underperformed standalone ResNet50 by seven points, despite having access to strictly more raw information in the form of combined deep features and pose descriptors. The most likely cause is that the PCA dimensionality reduction and mutual-information-based selection required to make the fused vector tractable for XGBoost discard more discriminative content than they retain, in contrast to allowing the convolutional network to learn a task-adapted representation through end-to-end optimisation.

F. ConvNeXt-Tiny with Data Augmentation

The strongest result in this study was achieved by ConvNeXt-Tiny trained on the augmented 2,729-image dataset: 80.55% validation accuracy, with both weighted precision and recall at 81% and a weighted F1-score of 0.80. Table XI presents the per-class detail, and Fig. 7 displays the associated confusion matrix.

Class	Precision	Recall	F1-Score	Support
Foetus	0.71	0.88	0.79	98
Log	0.84	0.83	0.83	164
Starfish	0.85	0.70	0.77	88
Unknown	0.79	0.72	0.75	95
Yearner	0.84	0.87	0.86	100

TABLE XI. CONVNEXT-TINY PER-CLASS METRICS (AUGMENTED DATASET)

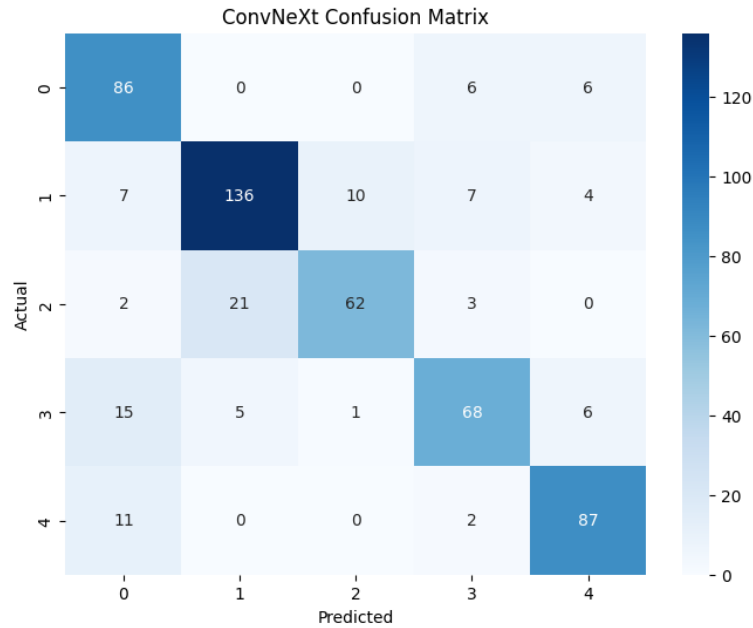


Fig. 7. ConvNeXt-Tiny confusion matrix on the augmented validation split (class order: Foetus, Log, Starfish, Unknown, Yearner).

ConvNeXt-Tiny correctly identified 86 out of 98 Foetus images, 136 out of 164 Log images, 62 out of 88 Starfish images, 68 out of 95 Unknown images, and 87 out of 100 Yearner images. Starfish remains the most challenging class for this model, most frequently being confused with Log (21 instances) and occasionally with Unknown (4 instances). This pattern is expected given the visual similarity between an extended Starfish posture and an elongated Log posture in infrared imagery, and the same confusion trend is observable, at reduced frequency, in the EfficientNetB3 confusion matrix in Fig. 6, indicating this reflects an inherent characteristic of the imaging data rather than a model-specific artefact.

Fig. 8 tracks the progression of training and validation accuracy and loss across the eight training epochs. Training accuracy increases steadily throughout, climbing from 56.0% to 93.8%, while validation accuracy advances more gradually, reaching its maximum at epoch 5 (80.55%) before declining to 76.3% at epoch 6 and only partially recovering over the final two epochs. Validation loss attains its minimum at the same epoch-5 checkpoint (0.538) and begins to rise thereafter even as training loss continues to decrease. This post-epoch-5 divergence is characteristic of mild overfitting, and the early-stopping criterion (patience 5 with respect to validation accuracy) had not yet activated by the point training concluded at epoch 8. This should be borne in mind when assessing how confidently the reported 80.55% can be considered a reliable performance ceiling rather than an optimistic single snapshot.

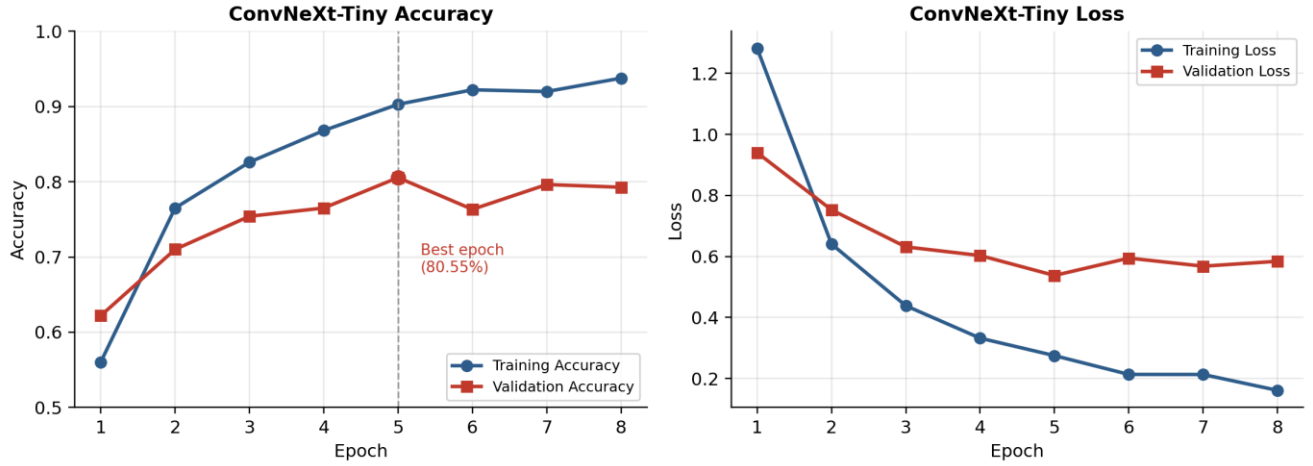


Fig. 8. ConvNeXt-Tiny training/validation accuracy and loss across 8 epochs. Validation accuracy peaks at epoch 5 (80.55%) while training accuracy keeps climbing, indicating mild overfitting beyond that point.

G. Overall Discussion

Viewed together, the nine outcomes in Table V trace a largely orderly progression. Manually engineered joint geometry alone proves insufficiently expressive for reliably separating five sleep postures, and automated feature selection does not alter this conclusion, since the performance ceiling is governed by the information content of the features rather than their quantity; this negative finding is precisely what motivates treating the Q-Learning search of Algorithm 1 as a diagnostic instrument rather than purely an optimisation step. A small but well-selected set of pose measurements outperforms a substantially larger and more elaborately constructed manual feature set, a practical reminder that greater feature engineering effort does not translate automatically into better representations when training samples are limited. Allowing a convolutional network to acquire its own representation directly from image pixels surpasses every keypoint-based strategy by a substantial margin, and augmenting the learned representation with manually constructed pose features did not improve matters; if anything, the dimensionality reduction required to render the combined vector tractable sacrificed more discriminative content than the additional features contributed. Finally, coupling a contemporary efficient backbone with aggressive data augmentation yielded the highest performance overall, although the present study cannot cleanly disentangle the contribution of the enlarged training set from that of the backbone architecture, since neither an unaugmented ConvNeXt-Tiny run nor augmented ResNet50 and EfficientNetB3 runs fell within scope. The relatively small original dataset (455 images) and the consequent fold-to-fold performance variability—most apparent in the 15-point range in Table X—should also be considered when extrapolating these precise figures to other datasets or imaging setups.

V. Conclusion

This investigation compared six approaches to representing infrared sleep-posture images for classification: manually engineered joint geometry, a novel Q-Learning-selected subset of that geometry (Algorithm 1), a compact pose-only descriptor, deep transfer learning, a combined deep-pose architecture, and an augmented ConvNeXt-Tiny model, all assessed under a single unified evaluation protocol. Classification performance improved progressively along this sequence, from 21.5%–29.0% for the manually crafted and reinforcement-learning-selected features, to 42.3%–43.3% for the pose-only descriptors, to 62.0%–63.7% for direct deep transfer learning, with a regression to 56.7% for naive deep-pose fusion, and finally reaching 80.55% (weighted F1-score of 0.80) for ConvNeXt-Tiny trained on the augmented dataset—the best-performing configuration in the study.

Three natural directions for future work emerge from these results. Expanding the IRS dataset to include a larger, more balanced collection would clarify whether the performance gap between manual and learned representations narrows with increasing sample size, and would also permit isolated comparisons of backbone effects versus augmentation effects through unaugmented ConvNeXt-Tiny and augmented ResNet50/EfficientNetB3 experiments. Constructing an ensemble from ResNet50 and EfficientNetB3 is another promising avenue, given the complementary error profiles the two models exhibit across classes. Finally, extending the processing pipeline to incorporate temporally successive overnight frames and integrating physiological measurements such as heart rate could advance this line of research beyond single-frame posture recognition toward a more comprehensive assessment of sleep quality and stress.

References

- [1] L. Rachakonda, S. P. Mohanty, E. Kougianos, K. B. Karunakaran, and M. K. Ganapathiraju, "Smart-pillow: An IoT based device for stress detection considering sleeping habits," 2018 IEEE International Symposium on Smart Electronic Systems (iSES), pp. 161–166, 2018.
- [2] Y. Zhang, A. Xiao, T. Zheng, H. Xiao, and R. Huang, "The relationship between sleeping position and sleep quality: A flexible sensor-based study," *Sensors*, vol. 22, p. 6220, 2022.
- [3] R. K. Nath, H. Thapliyal, A. Caban-Holt, and S. P. Mohanty, "Machine learning based solutions for real-time stress monitoring," *IEEE Consumer Electronics Magazine*, vol. 9, no. 5, pp. 34–41, 2020.
- [4] L. Rachakonda, A. Bapatla, S. Mohanty, and E. Kougianos, "Sayopillow: Blockchain-integrated privacy-assured IoMT framework for stress management considering sleeping habits," *IEEE Transactions on Consumer Electronics*, vol. PP, pp. 1–1, 2020.
- [5] F. Akhtar, M. B. Bin Heyat, J. P. Li, P. K. Patel, Rishipal, and B. Guragai, "Role of machine learning in human stress: A review," in 2020 17th Int. Computer Conf. on Wavelet Active Media Technology and Information Processing (ICCWAMTIP), 2020, pp. 170–174.
- [6] M. M. Rahman, A. Mohaimenul Islam, J. Miah, S. Ahmad, and M. Mamun, "Sleepwell: Stress level prediction through sleep data. Are you stressed?" in 2023 IEEE World AI IoT Congress (AllIoT), 2023, pp. 0229–0235.
- [7] K. Tang, A. Kumar, M. Nadeem, and I. Maaz, "CNN-based smart sleep posture recognition system," *IoT*, vol. 2, no. 1, pp. 119–139, 2021.
- [8] S. Dhamchatsoontree, C. Sirisin, M. Proncharoensukkul, and K. Wongpatikaseree, "i-sleep: Intelligent sleep detection system for analyzing sleep behavior," in 2019 4th Int. Conf. on Information Technology (InCIT), 2019, pp. 144–148.
- [9] J. G. Jayawickrama and R. A. H. M. Rupasingha, "Human stress detection based on sleeping habits using machine learning algorithms," in 2023 3rd Int. Conf. on Advanced Research in Computing (ICARC), 2023, pp. 7–12.
- [10] P. Barsocchi, M. Bianchini, A. Crivello, D. La Rosa, F. Palumbo, and F. Scarselli, "An unobtrusive sleep monitoring system for the human sleep behaviour understanding," in 2016 7th IEEE Int. Conf. on Cognitive Infocommunications (CogInfoCom), 2016, pp. 91–96.
- [11] R. K. Nath, H. Thapliyal, A. Caban-Holt, and S. P. Mohanty, "Machine learning based solutions for real-time stress monitoring," *IEEE Consumer Electronics Magazine*, vol. 9, no. 5, pp. 34–41, 2020.
- [12] A. Kapoor and S. Goel, "Predicting stress at workplace using machine learning techniques," in 2023 14th Int. Conf. on Computing Communication and Networking Technologies (ICCCNT), 2023, pp. 1–5.
- [13] J. G. Jayawickrama and R. Rupasingha, "Ensemble learning approach to human stress detection based on behaviours during the sleep," in 2022 4th Int. Conf. on Advancements in Computing (ICAC), 2022, pp. 132–137.
- [14] A. Muaremi, A. Bexheti, F. Gravenhorst, B. Amrich, and G. Tröster, "Monitoring the impact of stress on the sleep patterns of pilgrims using wearable sensors," in IEEE-EMBS Int. Conf. on Biomedical and Health Informatics (BHI), 2014, pp. 185–188.
- [15] G. Bradski, "The OpenCV Library," *Dr. Dobb's Journal of Software Tools*, 2000.
- [16] C. Lugaresi, J. Tang, H. Nash, C. McClanahan, E. Uboweja, M. Hays, F. Zhang, C.-L. Chang, M. G. Yong, J. Lee, W.-T. Chang, W. Hua, M. Georg, and M. Grundmann, "Mediapipe: A framework for building perception pipelines," *arXiv:1906.08172*, 2019.
- [17] C. Zheng, W. Wu, T. Yang, S. Zhu, C. Chen, R. Liu, J. Shen, N. Kehtarnavaz, and M. Shah, "Deep learning-based human pose estimation: A survey," *ACM Computing Surveys*, vol. 56, pp. 1–37, 2020.
- [18] N. Febriana, A. Rizal, and E. Susanto, "Sleep monitoring system based on body posture movement using Microsoft Kinect sensor," vol. 2092, 2019.
- [19] D. S. Elvitigala, D. J. C. Matthies, and S. Nanayakkara, "Stressfoot: Uncovering the potential of the foot for acute stress sensing in sitting posture," *Sensors*, vol. 20, no. 10, 2020.
- [20] B. C. Galland, G. Reeves, B. J. Taylor, and D. P. G. Bolton, "Sleep position, autonomic function, and arousal," *Archives of Disease in Childhood – Fetal and Neonatal Edition*, vol. 78, no. 3, pp. F189–F194, 1998.